

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch: Civil Semester: VI Subject Code & Name: BTCVC601 & Design of Concrete Structures I Max Marks: 60 Date: Duration: 3 Hr.			
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks	
Q. 1	Solve Any Two of the following. (This is just a sample instruction)			
A)	Explain the stress -strain behavior of steel and concrete.	CO - 1	6	
B)	State and explain the different load combinations.	CO - 1	6	
C)	Explain with neat sketch the balanced section, under reinforced section and over reinforced section.	CO - 3	6	
Q.2	Solve Any Two of the following. (This is just a sample instruction)			
A)	Determine the moment of resistance of a singly reinforced beam of 160 mm wide and 300 mm deep up to the center of reinforcement. If the stresses in steel and concrete are not exceed 140 N/mm ² and 5 N/mm ² . The reinforcement consists of 4 bars of 16 mm diameter. Take m = 18. If the above beam is used over an effective span of 5 m. Find the maximum load the beam can carry, Inclusive of its own weight.	CO - 3	6	
B)	Design a simply supported slab supported on masonry walls to the following requirements: i. Clear Span – 3 m. ii. Live Load – 2.5 KN/m ² . iii. Floor Finish Load – 1.5 KN/m ² . iv. Thickness of masonry Wall – 120 mm at each end. Use M - 20 concrete and Fe – 415 steels.	CO - 2	6	
C)	Explain the types of footing and its function.	CO - 4	6	
Q. 3	Solve Any Two of the following. (This is just a sample instruction)			
A)	Determine maximum axial working load that can be carried by 230 mm X 450 mm rectangular column reinforced with 8 no's of 16 mm diameter bars of Fe – 415 steel. One end of the column is fixed and another end is hinged. The length of the column is 3.2 m. Concrete grade used is M – 20. Check the column for minimum eccentricity also.	CO - 4	6	

B)	Design a dog legged RC stair of room size 3.5 m X 6.17 m with following data. Vertical distance between the floors is 3.75 m, Width of stair = 1.6 m, Tread = 270 mm (11 Nos. of Trades up to mid landing), Rise = 156.2 mm (12 Nos. of Risers up to mid landing), Mid Landing = 1.6 m, Live load = 2 KN/m ² , Wall Thickness 230 mm. Use M - 20 Concrete and Fe – 415 Steel.	CO - 2	6
C)	Explain the Terms i. Characteristic Strength ii. Characteristic Load iii. Partial Safety Factor	CO - 1	6
Q.4	Solve Any Two of the following. <i>(This is just a sample instruction)</i>		
A)	Explain the “Limit State Method” and its types.	CO - 1	6
B)	Why the doubly reinforced section is used?	CO - 3	6
C)	A RCC T beam having a total depth of 400 mm, width 250 mm is cast monolithically with slab of 120 mm thick. The effective width of the slab for the beam is 1770 mm, Concrete mix is of grade M – 20 and steel of Fe – 500 have been used. Calculate Ultimate moment of Resistance if use 4 no’s of 25 mm diameter bar.	CO - 3	6
Q. 5	Solve Any Two of the following. <i>(This is just a sample instruction)</i>		
A)	Define the bond state and explain the types of bond.	CO - 2	6
B)	A beam has an effective depth of 400 mm and width 350 mm it consists of 4 bars of 25 mm diameter. The beam has a effective span of 5 m and carry’s a UDL of 90 KN/m over entire span. Design the shear reinforcement. The grade of concrete is M – 20 and Fe – 415 Steel.	CO - 3	6
C)	Find development length of 16 mm dia. bar in tension & compression. Use M20 concrete & Fe 500 grade steel and Fe – 250 M.S. Steel. Take $\tau_{bd} = 1.2$ N/mm ² .	CO - 2	6
	*** End ***		

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